WEBRUM-03: Core Web Vitals

> **Series:** WEBRUM | **Notebook:** 3 of 8 | **Created:** March 2026 | **Last Updated:** 03/12/2026

Overview

Core Web Vitals (CWV) are Google's standardized metrics for measuring real-world user experience on the web. They focus on three pillars: loading performance, interactivity, and visual stability. Dynatrace captures these metrics via the RUM JavaScript agent using the browser's PerformanceObserver API, making them available for DQL analysis in Grail.

This notebook covers the three Core Web Vitals — LCP, INP, and CLS — how Dynatrace measures them, what good/poor thresholds look like, and how to track them over time with DQL queries.

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Prerequisites

Requirement	Details
Dynatrace Environment	SaaS with Grail enabled
RUM Enabled	Web applications with Core Web Vitals capture enabled
Browser Support	CWV are Chromium-based only (Chrome, Edge); Safari/Firefox have partial support
Permissions	`storage:events:read`, `storage:metrics:read`
Previous Notebook	WEBRUM-01: RUM Fundamentals

1. Understanding Core Web Vitals

Google defines three Core Web Vitals that reflect real user experience:

Metric	Full Name	Measures	Good	Needs Improvement	Poor
LCP	Largest Contentful Paint	Loading performance	≤ 2.5s	2.5s – 4.0s	> 4.0s
INP	Interaction to Next Paint	Interactivity responsiveness	≤ 200ms	200ms – 500ms	> 500ms
CLS	Cumulative Layout Shift	Visual stability	≤ 0.1	0.1 – 0.25	> 0.25

> **Note:** INP replaced FID (First Input Delay) as a Core Web Vital in March 2024. INP measures the latency of *all* interactions during a session, not just the first one.

How Dynatrace Captures CWV

The RUM JavaScript agent uses the browser's `PerformanceObserver` API to capture:

- **LCP** — Observed via `largest-contentful-paint` entry type
- **INP** — Calculated from `event` entry types (click, keypress, pointerdown)
- **CLS** — Accumulated from `layout-shift` entries (excluding user-initiated shifts)

These values are reported as part of the user action data and are available as metrics in Grail.

2. Largest Contentful Paint (LCP)

LCP measures the time from when the user initiates navigation to when the largest content element (image, text block, video) is rendered in the viewport. It answers: **"How quickly does the main content appear?"**

Common LCP Elements

Element Type	Example
`<img>`	Hero image, product photo
`<video>` poster	Video thumbnail
Block-level text	Heading, paragraph
CSS `background-image`	Banner background

Factors That Impact LCP

- Server response time (TTFB)
- Render-blocking resources (CSS, JS)
- Resource load time (images, fonts)
- Client-side rendering delays

```
// Average LCP across all applications in the last 24 hours
fetch dt.rum.user_action, from:-24h
| filter action.type == "Load"
| filter isNotNull(largest.contentful.paint)
| summarize avg_lcp = avg(largest.contentful.paint),
    p75_lcp = percentile(largest.contentful.paint, 75),
    p95_lcp = percentile(largest.contentful.paint, 95),
    sample_size = count(),
    by:{app.name}
| sort avg_lcp desc
```

```
// LCP distribution — classify into Good / Needs Improvement / Poor
fetch dt.rum.user_action, from:-24h
| filter action.type == "Load"
| filter isNotNull(largest.contentful.paint)
| fieldsAdd lcp_ms = toDouble(largest.contentful.paint) / 1000000.0
| fieldsAdd lcp_category = if(lcp_ms &lt;= 2500, "Good",
    else: if(lcp_ms &lt;= 4000, "Needs Improvement",
    else: "Poor"))
| summarize action_count = count(), by:{lcp_category}
| fieldsAdd total = sum(action_count)
| fieldsAdd percentage = round(toDouble(action_count) / toDouble(total) *
100.0, decimals: 1)
```

3. Interaction to Next Paint (INP)

INP measures the time from when a user interacts (click, tap, keypress) to when the browser renders the next frame reflecting that interaction. Unlike FID (which only measured the *first* interaction), INP considers *all* interactions and reports the worst one (approximated by the 98th percentile).

What INP Captures

1. **Input delay** — Time from user interaction to event handler execution
2. **Processing time** — Time to execute event handlers
3. **Presentation delay** — Time from handler completion to next paint

Common INP Bottlenecks

Bottleneck	Symptom	Fix
Long tasks on main thread	High input delay	Break up long JavaScript tasks
Expensive event handlers	High processing time	Debounce, use web workers
Forced layout/reflow	High presentation delay	Batch DOM reads/writes

```
// INP percentile analysis by application
fetch dt.rum.user_action, from:-24h
| filter isNotNull(interaction.to.next.paint)
| fieldsAdd inp_ms = toDouble(interaction.to.next.paint) / 1000000.0
| summarize avg_inp = avg(inp_ms),
    p75_inp = percentile(inp_ms, 75),
    p95_inp = percentile(inp_ms, 95),
    sample_size = count(),
    by:{app.name}
| sort p75_inp desc
```

```
// INP by page — identify the slowest-responding pages
fetch dt.rum.user_action, from:-24h
| filter isNotNull(interaction.to.next.paint)
| fieldsAdd inp_ms = toDouble(interaction.to.next.paint) / 1000000.0
| summarize avg_inp = avg(inp_ms),
    p75_inp = percentile(inp_ms, 75),
    action_count = count(),
    by:{action.name}
| filter action_count &gt; 10
| sort p75_inp desc
| limit 10
```

4. Cumulative Layout Shift (CLS)

CLS measures unexpected layout shifts during the entire lifespan of a page. A layout shift occurs when a visible element changes position from one rendered frame to the next without user interaction.

Common Causes of CLS

Cause	Example	Fix
Images without dimensions	Image loads and pushes content down	Set `width` and `height` attributes
Dynamically injected content	Cookie banner pushes page down	Reserve space with CSS
Web fonts	FOUT (flash of unstyled text) causes text reflow	Use `font-display: swap` with fallback metrics
Ads/embeds	Third-party content loads late	Use `<iframe>` with fixed dimensions

```
// CLS distribution — classify into Good / Needs Improvement / Poor
fetch dt.rum.user_action, from:-24h
| filter action.type == "Load"
| filter isNotNull(cumulative.layout.shift)
| fieldsAdd cls_category = if(cumulative.layout.shift &lt;= 0.1, "Good",
    else: if(cumulative.layout.shift &lt;= 0.25, "Needs Improvement",
    else: "Poor"))
| summarize action_count = count(), by:{cls_category}
```

```
// Worst CLS pages — pages with the most layout shifting
fetch dt.rum.user_action, from:-24h
| filter action.type == "Load"
| filter isNotNull(cumulative.layout.shift)
| summarize avg_cls = avg(cumulative.layout.shift),
    p75_cls = percentile(cumulative.layout.shift, 75),
    action_count = count(),
    by:{action.name}
| filter action_count &gt; 10
| sort p75_cls desc
| limit 10
```

5. CWV by Page Group

Aggregate Core Web Vitals by page to identify which pages need optimization:

```
// All three CWV metrics by page — comprehensive page health view
fetch dt.rum.user_action, from:-24h
| filter action.type == "Load"
| summarize
    p75_lcp_ms = percentile(toDouble(largest.contentful.paint) / 1000000.0,
75),
    p75_cls = percentile(cumulative.layout.shift, 75),
    p75_inp_ms = percentile(toDouble(interaction.to.next.paint) / 1000000.0,
75),
    page_views = count(),
    by:{action.name}
| filter page_views &gt; 20
| fieldsAdd lcp_status = if(p75_lcp_ms &lt;= 2500, "Good", else:
if(p75_lcp_ms &lt;= 4000, "NI", else: "Poor")),
    cls_status = if(p75_cls &lt;= 0.1, "Good", else: if(p75_cls &lt;= 0.25,
"NI", else: "Poor")),
    inp_status = if(p75_inp_ms &lt;= 200, "Good", else: if(p75_inp_ms &lt;=
500, "NI", else: "Poor"))
| sort page_views desc
| limit 15
```

6. CWV Trends Over Time

Tracking CWV over time helps identify regressions after deployments, seasonal patterns, and the impact of optimizations.

```
// LCP trend over the last 7 days — hourly p75
fetch dt.rum.user_action, from:-7d
| filter action.type == "Load"
| filter isNotNull(largest.contentful.paint)
| fieldsAdd lcp_ms = toDouble(largest.contentful.paint) / 1000000.0
| makeTimeseries p75_lcp = percentile(lcp_ms, 75), interval:1h
```

```
// CLS trend over the last 7 days — daily p75
fetch dt.rum.user_action, from:-7d
| filter action.type == "Load"
| filter isNotNull(cumulative.layout.shift)
| makeTimeseries p75_cls = percentile(cumulative.layout.shift, 75),
interval:1d
```

7. CWV Scoring Dashboard

Create a single-query CWV scorecard showing the percentage of page loads in each category:

```
// CWV scorecard — percentage of Good / NI / Poor for each metric
fetch dt.rum.user_action, from:-24h
| filter action.type == "Load"
| fieldsAdd lcp_ms = toDouble(largest.contentful.paint) / 1000000.0,
    inp_ms = toDouble(interaction.to.next.paint) / 1000000.0
| summarize total = count(),
    lcp_good = countIf(lcp_ms &lt;= 2500),
    lcp_poor = countIf(lcp_ms &gt; 4000),
    cls_good = countIf(cumulative.layout.shift &lt;= 0.1),
    cls_poor = countIf(cumulative.layout.shift &gt; 0.25),
    inp_good = countIf(inp_ms &lt;= 200),
    inp_poor = countIf(inp_ms &gt; 500)
| fieldsAdd lcp_good_pct = round(toDouble(lcp_good) / toDouble(total) *
100.0, decimals: 1),
    lcp_poor_pct = round(toDouble(lcp_poor) / toDouble(total) * 100.0,
decimals: 1),
    cls_good_pct = round(toDouble(cls_good) / toDouble(total) * 100.0,
decimals: 1),
    cls_poor_pct = round(toDouble(cls_poor) / toDouble(total) * 100.0,
decimals: 1),
    inp_good_pct = round(toDouble(inp_good) / toDouble(total) * 100.0,
decimals: 1),
    inp_poor_pct = round(toDouble(inp_poor) / toDouble(total) * 100.0,
decimals: 1)
```

8. Summary and Next Steps

In this notebook, we covered:

- **Core Web Vitals overview** — LCP, INP, and CLS with Google's thresholds
- **LCP analysis** — Loading performance measurement and distribution
- **INP analysis** — Interactivity measurement replacing FID
- **CLS analysis** — Visual stability scoring and worst pages
- **Page-level breakdown** — CWV per page group for targeted optimization
- **Trend tracking** — CWV over time for regression detection

- **Scoring dashboard** — Unified CWV scorecard query

Next Steps

- **WEBRUM-04: Session Analysis** — User journey mapping and conversion tracking
- **WEBRUM-06: Performance Analysis** — Deeper performance waterfall analysis beyond CWV

References

- Google Core Web Vitals
- Dynatrace Web Vitals
- INP Documentation
